

SSILP Math

Beginner

EXPONENTIAL FUNCTIONS



Learning objectives

Understand Exponential Functions and their graphs

Understand the graph of exponential functions with different bases

Compare and contrast exponential growth and decay

Apply Exponential Functions

Solve real-world problems involving exponential growth and decay, such as population growth and heat dissipation

Understand the Natural Exponential Function

Compare Exponential and Power Functions

Differentiate between exponential functions and power functions.

Session outline

1. Exponential Functions

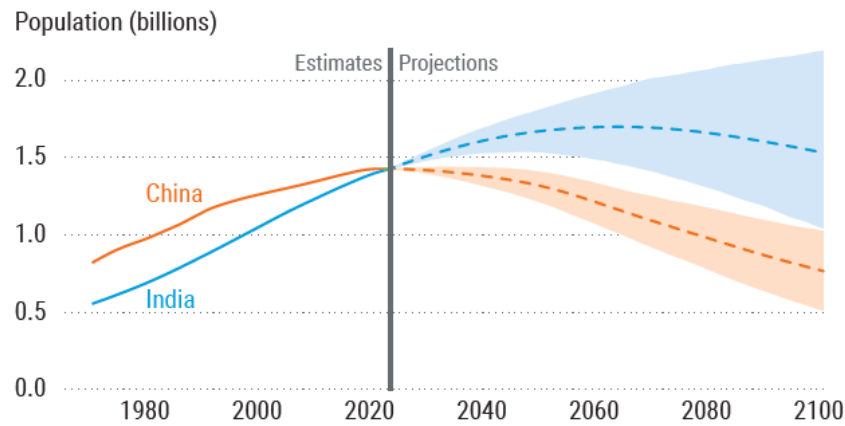
2. The natural exponential
function

1. Exponential Functions

India overtakes China as the world's most populous country

Figure 1

Trends in total population for China and India, estimates for 1970-2022 and projections for 2023-2100 (with 95 per cent prediction intervals)

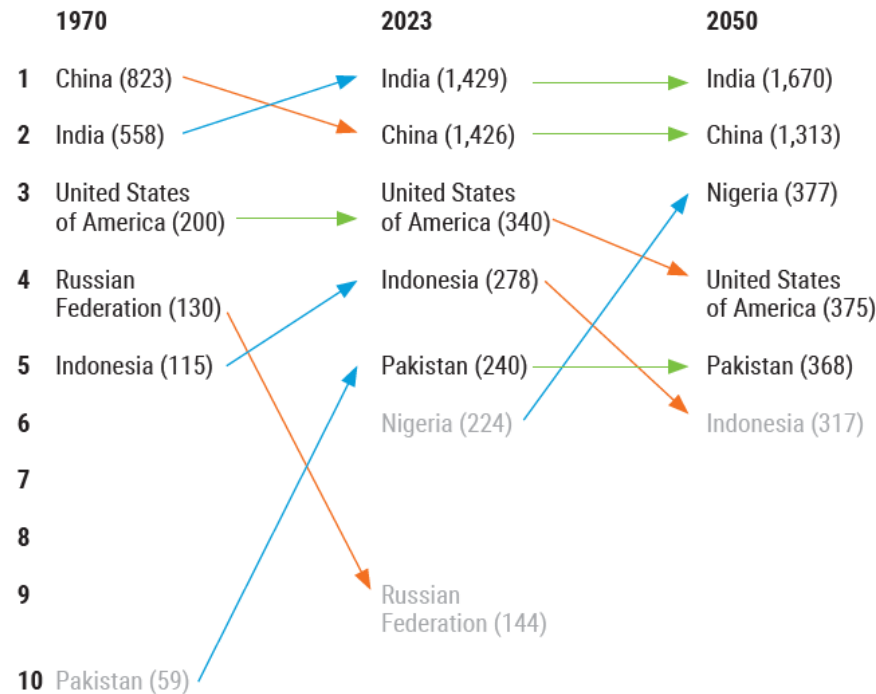


Data source: United Nations, *World Population Prospects 2022*, <https://population.un.org/wpp/>.

1 United Nations, Department of Economic and Social Affairs, Population Division (2022). *World Population Prospects 2022: Summary of Results*. UN DESA/POP/2022/TR/NO. 3.

Figure 2

Top five most populous countries, estimates for 1970 and projections for 2023 and 2050



Note: Numbers in parentheses refer to total population (in millions) on 1 July of the referenced year.

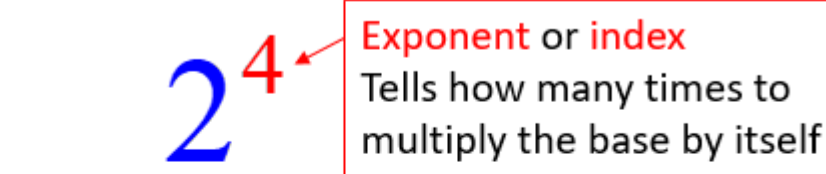
Data source: United Nations, *World Population Prospects 2022*, <https://population.un.org/wpp/>.

Two “population billionaires”,

China and India,

face **divergent demographic futures**

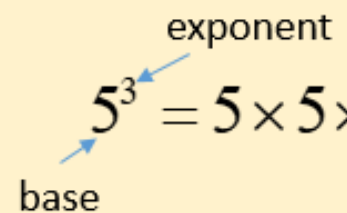
What is an exponent?



$2^4 = 2 \times 2 \times 2 \times 2 = 16$

Types of Exponents

Positive Exponent



$5^3 = 5 \times 5 \times 5$

Negative Exponent

$$a^{-x} = \frac{1}{a^x}$$

$$2^{-4} = \frac{1}{2^4}$$

Rational Exponent

$$a^{\frac{1}{n}} = \sqrt[n]{a}$$

$$a^{\frac{m}{n}} = \left(\sqrt[n]{a} \right)^m$$

Zero Exponent

$$a^0 = 1$$

What are Exponential functions?

$$y = 2^x$$

Variable in exponent

base

Examples of exponential functions

$$y = 10^x$$

$$y = 3^{(x-2)}$$

base > 0

base $\neq 1$

Exponential Functions: Formal Definition

EXPONENTIAL FUNCTIONS

The exponential function with base a is defined for all real numbers x by

$$f(x) = a^x$$

where $a > 0$ and $a \neq 1$.

We assume that $a \neq 1$ because the function $f(x) = 1^x = 1$ is just a constant function. Here are some examples of exponential functions:

$$f(x) = 2^x$$

Base 2

$$g(x) = 3^x$$

Base 3

$$h(x) = 10^x$$

Base 10

Evaluating exponential functions

$$y = 1.5(4^x)$$

evaluate for $x = 3$

$$y = 1.5(4^3)$$

Substitute $x=3$

$$y = 1.5(64)$$

Simplify the power first,
and then multiply

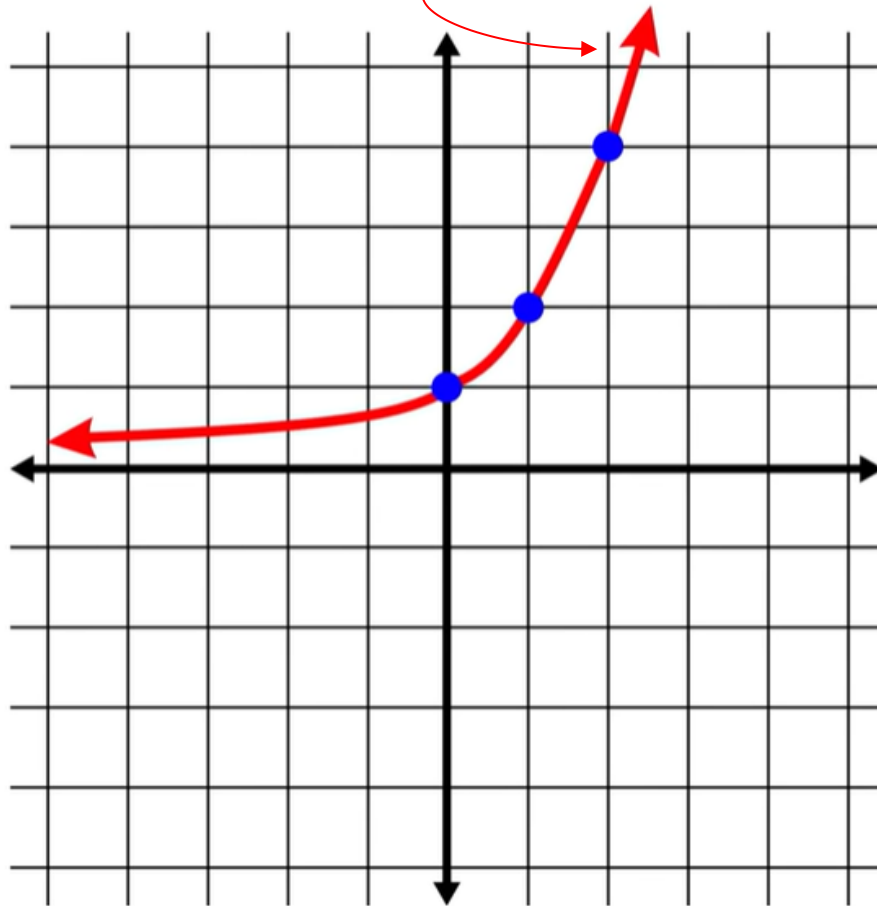
$$y = 96$$



Remember to follow
order of operations
correctly.

Graph of Exponential function (base>1)

the function grows
very rapidly as x
increases

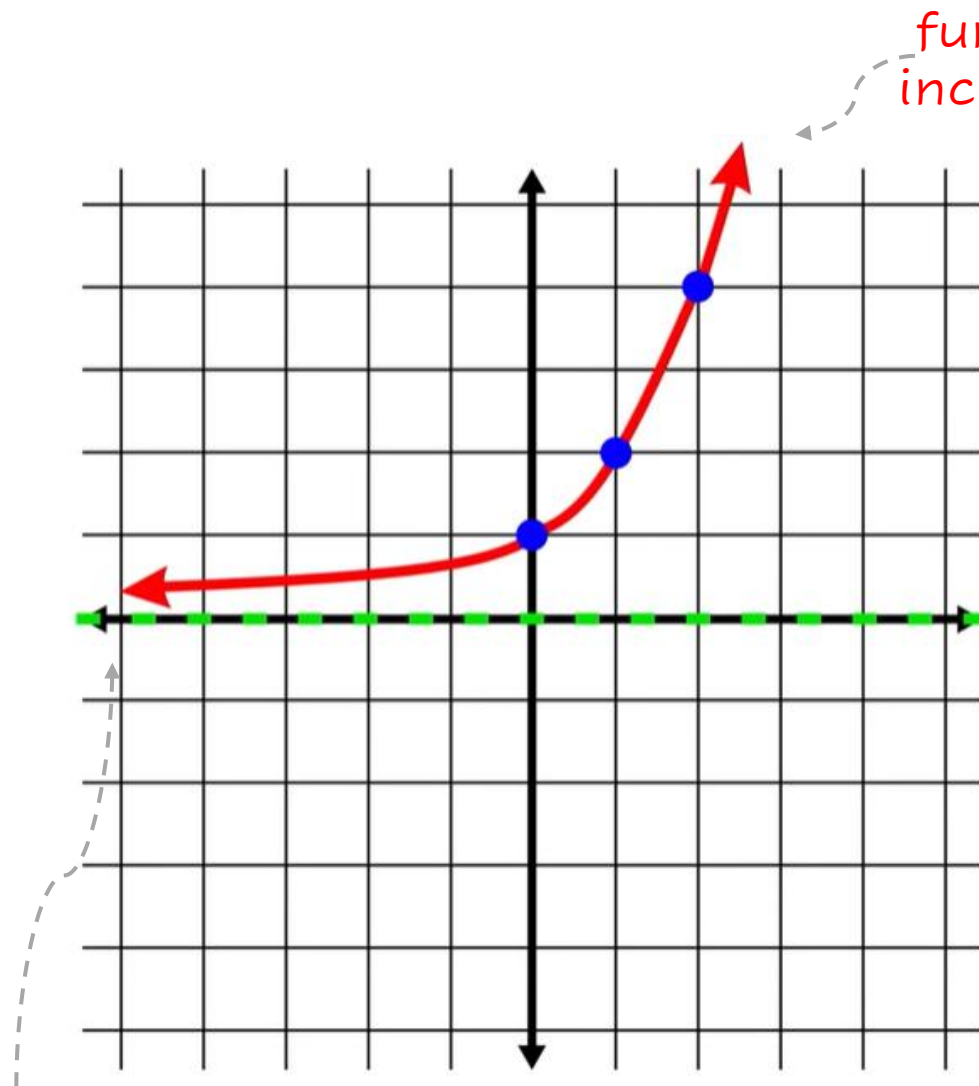


$$y = 2^x$$

x	y
0	$2^0 = 1$
1	$2^1 = 2$
2	$2^2 = 4$
3	$2^3 = 8$

This is an example of 'exponential growth'

Graph of Exponential function (base>1)



function approaches positive infinity as x increases

$$y = 2^x$$

horizontal asymptote: $y = 0$

the function is always **positive**
(raising to any exponent yields positive values)

$$2^{-1} = 1/2^1 = 1/2$$

$$2^{-2} = 1/2^2 = 1/4$$

$$2^{-3} = 1/2^3 = 1/8$$

$$2^{-4} = 1/2^4 = 1/16$$

getting closer to zero
as the exponent decreases

Graph of Exponential function, $0 < \text{base} < 1$

As the exponent becomes increasingly **negative**, the function approaches **positive infinity**.

$$y = \left(\frac{1}{2}\right)^x$$

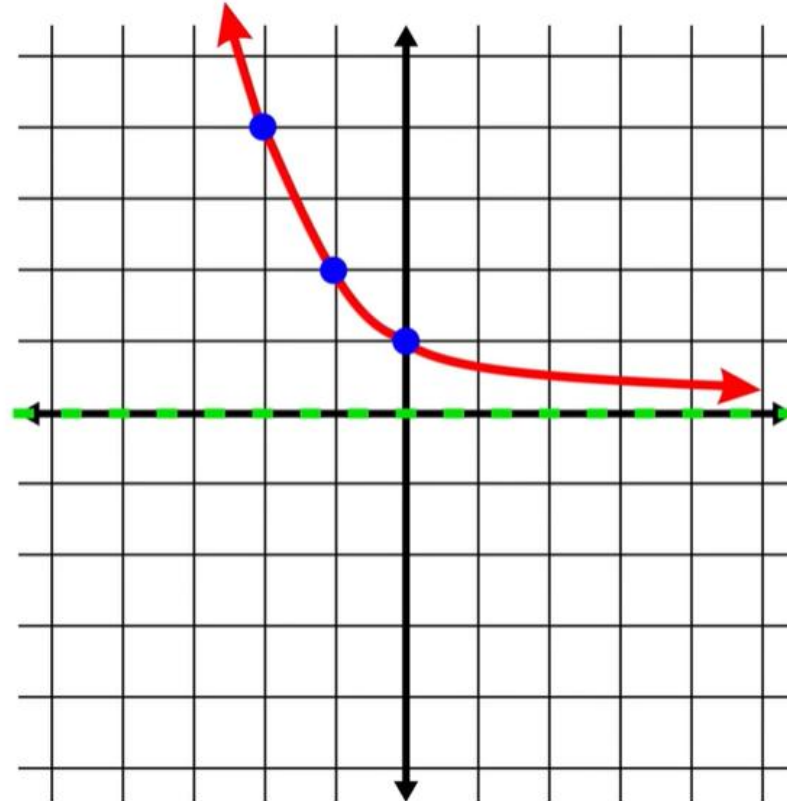
$$\left(\frac{1}{2}\right)^{-1} = 2^1 = 2$$

$$\left(\frac{1}{2}\right)^{-2} = 2^2 = 4$$

$$\left(\frac{1}{2}\right)^{-3} = 2^3 = 8$$

$$\left(\frac{1}{2}\right)^{-4} = 2^4 = 16$$

$$\left(\frac{1}{2}\right)^{-5} = 2^5 = 32$$



As the exponent **increases**, the function approaches **zero**,
→ **horizontal asymptote** in the X axis.

$$\left(\frac{1}{2}\right)^0 = 1$$

$$\left(\frac{1}{2}\right)^1 = \frac{1}{2}$$

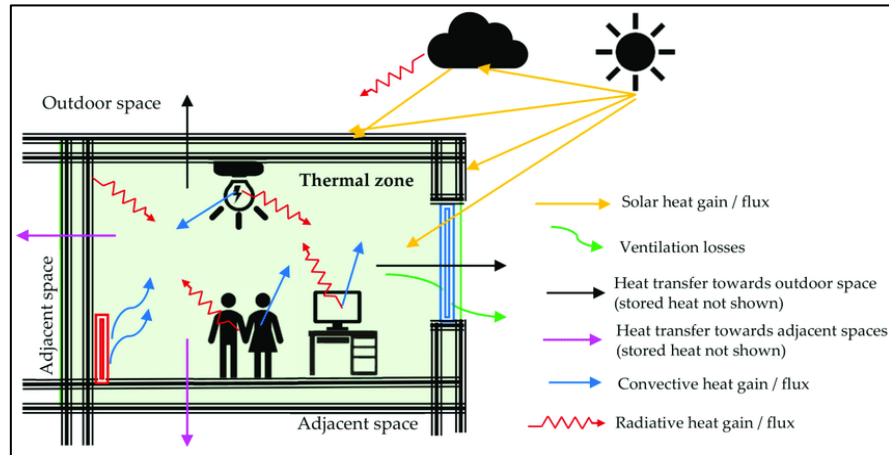
$$\left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

$$\left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

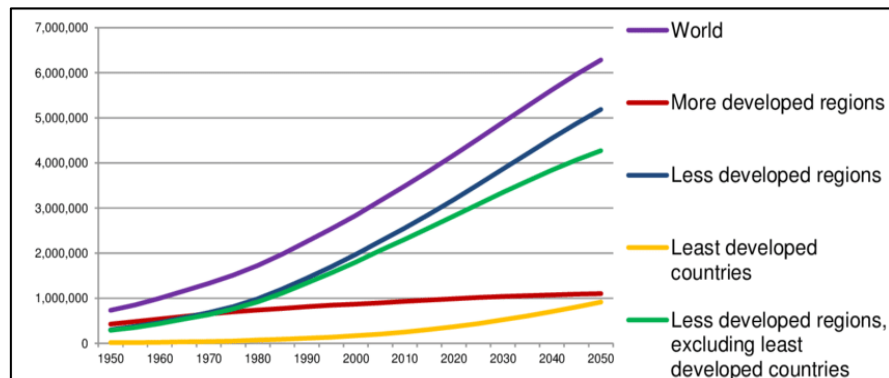
$$\left(\frac{1}{2}\right)^4 = \frac{1}{16}$$

This is an example of 'exponential decay'

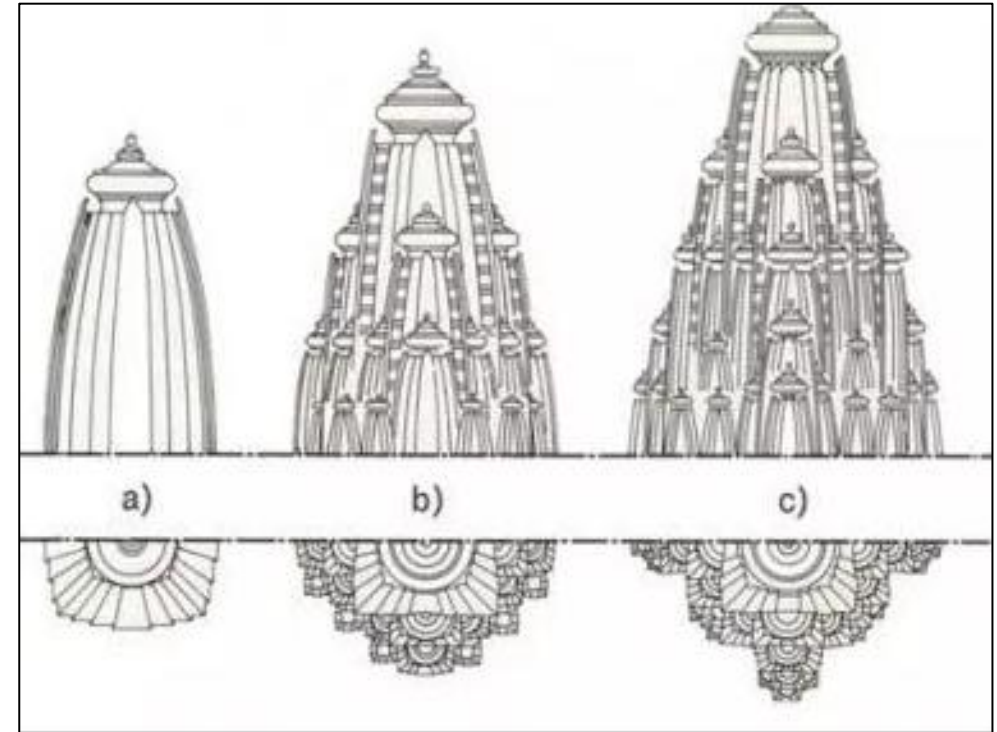
Applications of exponential functions



Exponential decay functions describe how **heat dissipates** over time and distance



predict population increases and the associated demands on infrastructure and resources.



Model **fractals**, used in various design elements like façades, flooring patterns, and landscape design

Which of the following are not exponential functions?

Which of the following equations are *not* exponential functions?

- $f(x) = 4^{3(x-2)}$
- $g(x) = x^3$
- $h(x) = \left(\frac{1}{3}\right)^x$
- $j(x) = (-2)^x$

ACTIVITY 1.1

1.1 Which of the following are not exponential functions?

Which of the following equations are *not* exponential functions?

- $f(x) = 4^{3(x-2)}$
- $g(x) = x^3$
- $h(x) = \left(\frac{1}{3}\right)^x$
- $j(x) = (-2)^x$

By definition, an exponential function has a constant base and a variable exponent. Thus, $g(x) = x^3$ does not represent an exponential function because the base is variable and the exponent is constant.

Recall that the base b of an exponential function is always a positive constant, and $b \neq 1$. Thus, $j(x) = (-2)^x$ does not represent an exponential function because the base, -2 , is less than 0.

$f(x)$ and $h(x)$ are exponential functions.

Evaluating Exponential functions

1) $f(x) = 2^x; x = 3$

2) $g(x) = 5(3)^{x+1}; x = 3$

3) $h(x) = 6^{-x}; x = -2$

4) $f(x) = 2 \left(\frac{1}{2}\right)^{x-1}$ when $x = 5$

ACTIVITY 1.2

1.2 Evaluating Exponential functions

1) 8

2) 405

3) 36

4) $f(x) = 2 \left(\frac{1}{2}\right)^{x-1}$

$$f(x) = 2 \left(\frac{1}{2}\right)^{x-1}$$

$$f(5) = 2 \left(\frac{1}{2}\right)^{5-1}$$

$$f(5) = 2 \left(\frac{1}{16}\right) = \frac{1}{8}$$

Graphs of Exponential Functions

Draw the graph of each function.

(a) $f(x) = 3^x$ (b) $g(x) = \left(\frac{1}{3}\right)^x$

SOLUTION We calculate values of $f(x)$ and $g(x)$ and plot points to sketch the graphs in Figure 1.

x	$f(x) = 3^x$	$g(x) = \left(\frac{1}{3}\right)^x$
-3	$\frac{1}{27}$	27
-2	$\frac{1}{9}$	9
-1	$\frac{1}{3}$	3
0	1	1
1	3	$\frac{1}{3}$
2	9	$\frac{1}{9}$
3	27	$\frac{1}{27}$

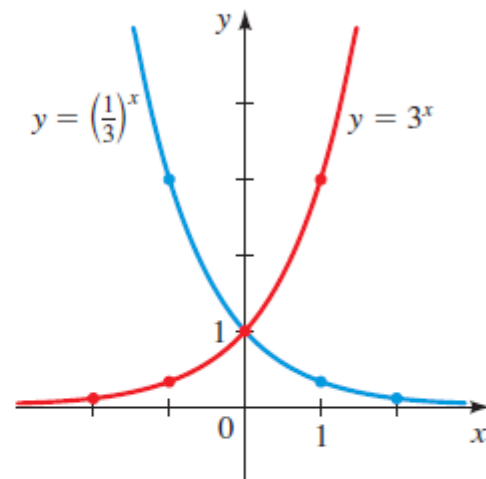


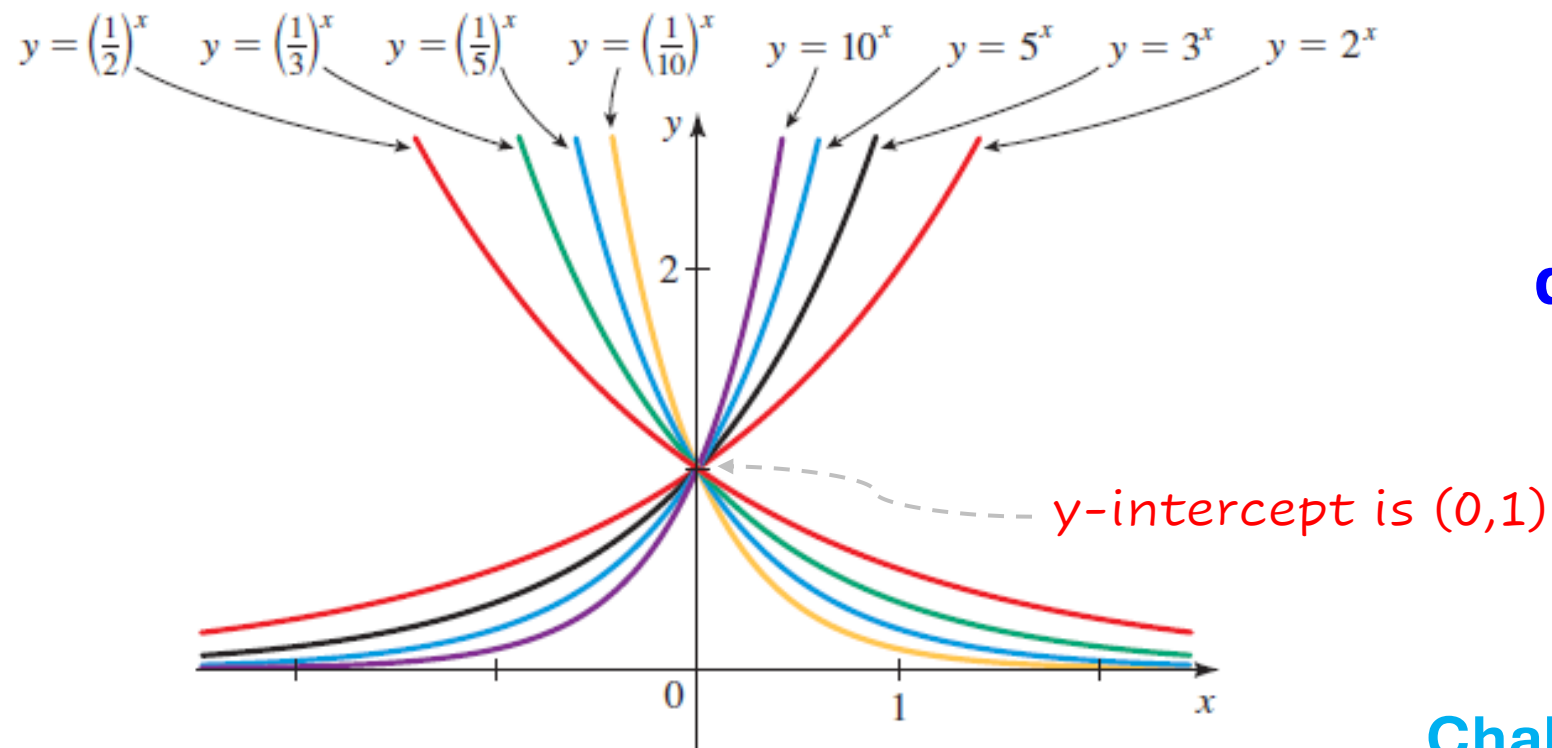
FIGURE 1

Notice that

$$g(x) = \left(\frac{1}{3}\right)^x = \frac{1}{3^x} = 3^{-x} = f(-x)$$

so we could have obtained the graph of g from the graph of f by reflecting in the y -axis.

Family of Exponential functions



changing the **base**
will produce
different exponential
functions

Figure 2 shows the graphs of the family of exponential functions $f(x) = a^x$ for various values of the base a . All of these graphs pass through the point $(0, 1)$ because $a^0 = 1$ for $a \neq 0$. You can see from Figure 2 that there are two kinds of exponential functions: If $0 < a < 1$, the exponential function decreases rapidly. If $a > 1$, the function increases rapidly (see the margin note).

Challenge:
Can you fold an A4 paper in
half 8 times?

Graphs of Exponential functions

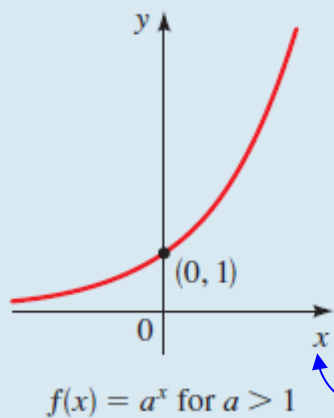
The x -axis is a horizontal asymptote for the exponential function $f(x) = a^x$. This is because when $a > 1$, we have $a^x \rightarrow 0$ as $x \rightarrow -\infty$, and when $0 < a < 1$, we have $a^x \rightarrow 0$ as $x \rightarrow \infty$ (see Figure 2). Also, $a^x > 0$ for all $x \in \mathbb{R}$, so the function $f(x) = a^x$ has domain $\mathbb{R}$ and range $(0, \infty)$. These observations are summarized in the following box.

GRAPHS OF EXPONENTIAL FUNCTIONS

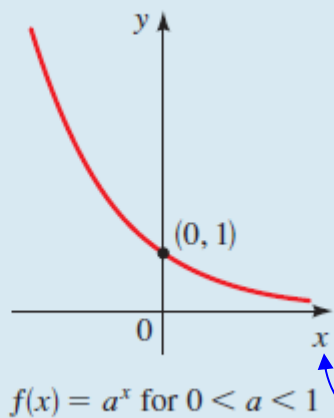
The exponential function

$$f(x) = a^x \quad a > 0, a \neq 1$$

has domain $\mathbb{R}$ and range $(0, \infty)$. The line $y = 0$ (the x -axis) is a horizontal asymptote of f . The graph of f has one of the following shapes.



Function rises to the right



Function dips down to the right

domain:
all real numbers

range:
all positive
real numbers

Comparing Exponential functions and Power functions

power function has its
variable in its base

POWER FUNCTIONS

$$y=x^1 \text{ (linear)}$$

$$y=x^2 \text{ (quadratic)}$$

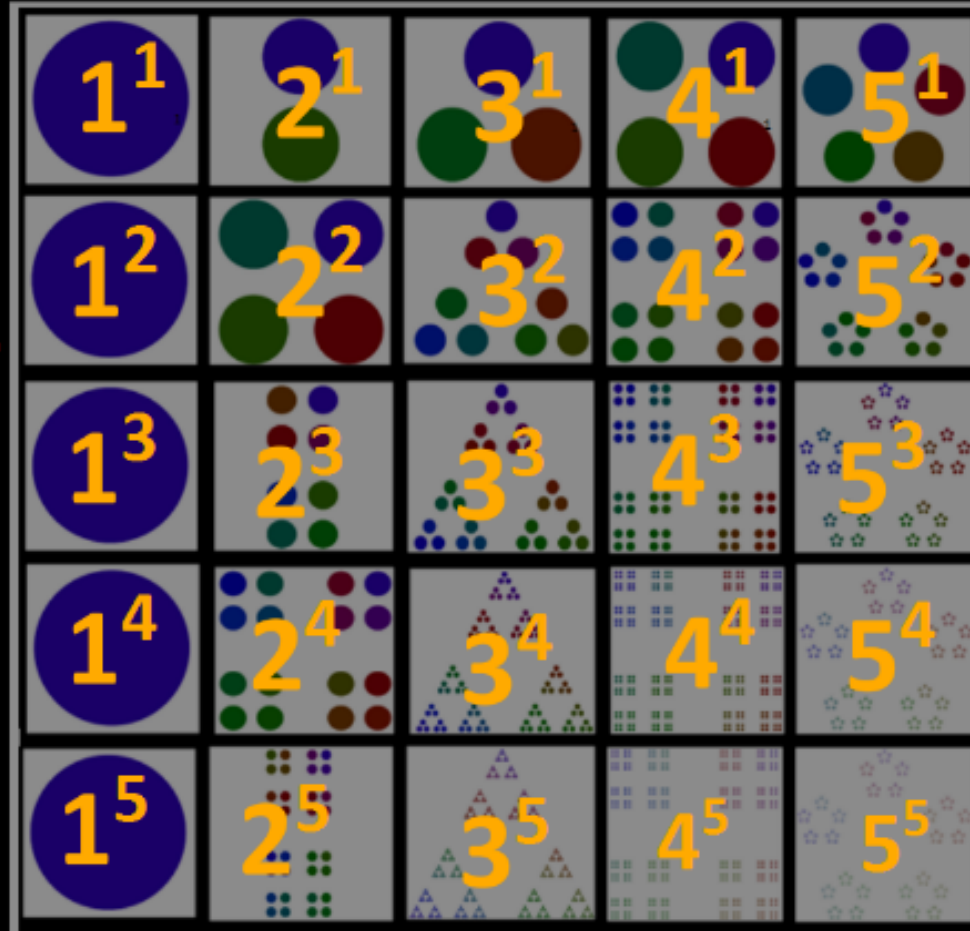
$$y=x^3 \text{ (cubic)}$$

$$y=x^4 \text{ (quartic)}$$

$$y=x^5$$

EXPONENTIAL FUNCTIONS

$$y=1^x \quad y=2^x \quad y=3^x \quad y=4^x \quad y=5^x$$



an exponential function
has its **variable in its
exponent**

An exponential
function **increases
faster** for larger x-
values

Graphing Exponential Functions

Graph both functions on one set of axes.

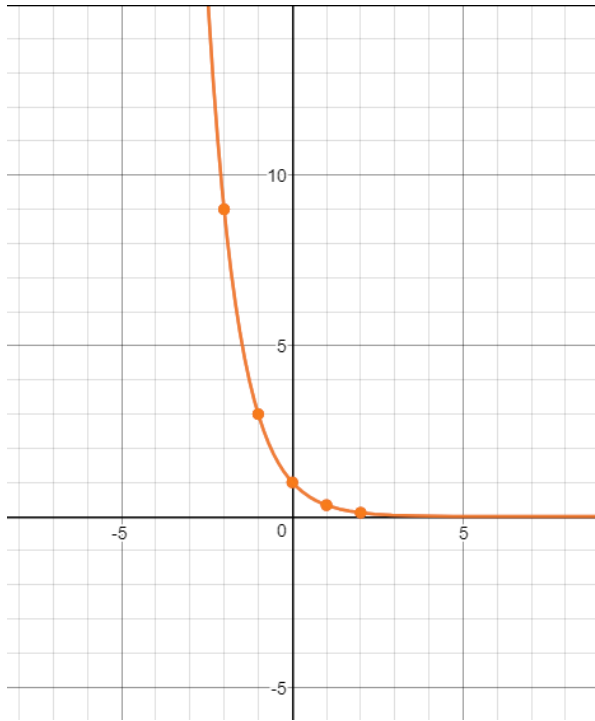
$$f(x) = 3^{-x} \quad \text{and} \quad g(x) = \left(\frac{1}{3}\right)^x$$

ACTIVITY 1.3

1.3 Graphing exponential functions

$$f(x) = 3^{-x} \quad \text{and} \quad g(x) = \left(\frac{1}{3}\right)^x$$

The two functions have exactly the same graphs since $3^{-x} = \frac{1}{3^x}$.



(1) **Bacteria Growth** A bacteria culture contains 1500 bacteria initially and doubles every hour.

- (a) Find a function N that models the number of bacteria after t hours.
- (b) Find the number of bacteria after 24 hours.

(2) **Mouse Population** A certain breed of mouse was introduced onto a small island with an initial population of 320 mice, and scientists estimate that the mouse population is doubling every year.

- (a) Find a function N that models the number of mice after t years.
- (b) Estimate the mouse population after 8 years.

1.4 Word problems

1 (a)

After 1 hour, there are $1500 \times 2 = 3000$ bacteria.

After 2 hours, there are $(1500 \times 2) \times 2 = 6000$ bacteria.

After 3 hours, there are $((1500 \times 2) \times 2) \times 2 = 12,000$ bacteria.

We see that after t hours, there are $N(t) = 1500 (2^t)$ bacteria.

1 (b)

After 24 hours, there are $N(24) = 1500 (2^{24}) = 25,165,824,000$ bacteria

1.4 Word problems

2

(a)

Because the population doubles every year, there are $N(t) = 320 (2^t)$ mice after t years.

(b)

After 8 years, there are approximately $N(8) = 320(2^8) = 81,920$ mice.

SOLUTIONS

2. The natural exponential function

The number 'e'

Much like π and i , e is a special number used in Math.

It is discovered by a Mathematician **Leonhard Euler** (Sounds like 'Oiler')

It is called the **Natural Base e** or the **Euler Number**.

$$e \approx 2.7182$$

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n$$

The number e is defined as the value that

$$\left(1 + \frac{1}{n} \right)^n$$

approaches, as n becomes very large

Type equation here.

n	$\left(1 + \frac{1}{n} \right)^n$
1	2.00000
5	2.48832
10	2.59374
100	2.70481
1000	2.71692
10,000	2.71815
100,000	2.71827
1,000,000	2.71828

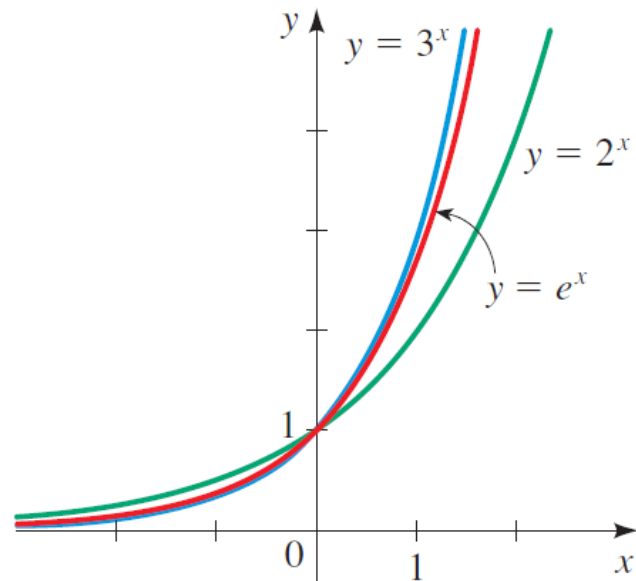
Natural Exponential Function

THE NATURAL EXPONENTIAL FUNCTION

The **natural exponential function** is the exponential function

$$f(x) = e^x$$

with base e . It is often referred to as *the* exponential function.



Graph of the natural exponential function

Since $2 < e < 3$,
the graph of the natural exponential function
lies between
the graphs of $y = 2^x$ and $y = 3^x$

Linear growth vs Exponential growth

LINEAR GROWTH

Consider the relationship represented by this table:

x	12	15	18	21
y	-2	5	12	19

Here, the x -values increase by exactly 3 units each time,

x		$\curvearrowright +3$		$\curvearrowright +3$		$\curvearrowright +3$	
	12		15		18		21

and the y -values increase by a constant *difference* of 7.

y		$\curvearrowright +7$		$\curvearrowright +7$		$\curvearrowright +7$	
	-2		5		12		19

Therefore, this relationship is *linear* because each y -value is 7 *more than* the value before it.

EXPONENTIAL GROWTH

Consider the relationship represented by this table:

x	0	1	2	3
y	1	3	9	27

Here, the x -values increase by exactly 1 unit each time,

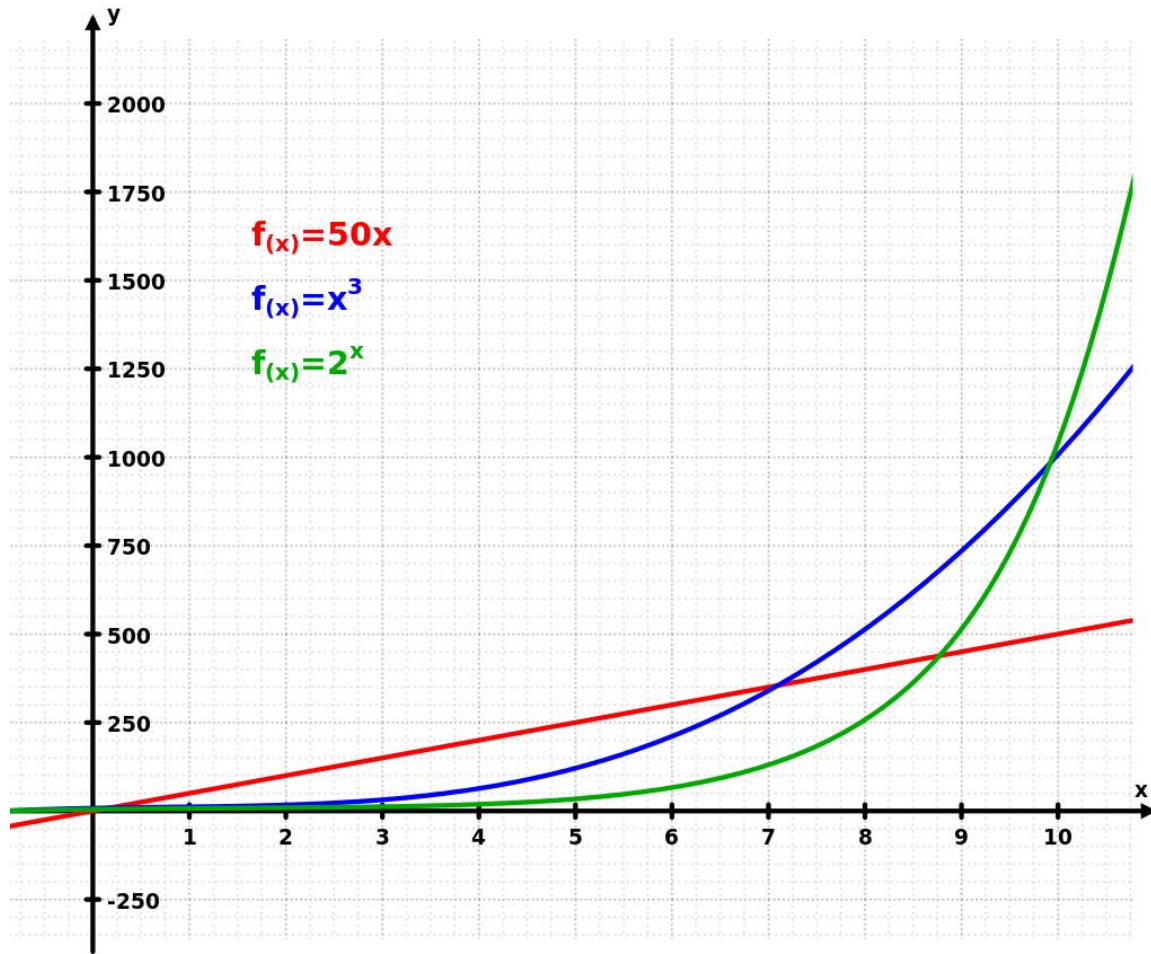
x		$\curvearrowright +1$		$\curvearrowright +1$		$\curvearrowright +1$	
	0		1		2		3

and the y -values increase by a constant *factor* of 3.

y		$\curvearrowright \times 3$		$\curvearrowright \times 3$		$\curvearrowright \times 3$	
	1		3		9		27

Therefore, this relationship is *exponential* because each y -value is 3 *times* the value before it.

Exponential growth



exponential growth surpasses both linear and cubic growth.

Evaluate Exponential functions with base e

a. $f(x) = e^x; x = 3$

b. $g(x) = 5(e)^{x+1}; x = 3$

c. $h(x) = e^{-x}; x = -2$

ACTIVITY 2.1

Exponential functions with base e

a.

$$f(x) = e^x; x = 3$$

$$f(3) = e^3 = 20.086$$

b.

$$g(x) = 5(e)^{x+1}; x = 3$$

$$\begin{aligned} g(3) &= 5(e)^{3+1} \\ &= 272.991 \end{aligned}$$

c.

$$h(x) = e^{-x}; x = -2$$

$$\begin{aligned} h(-2) &= e^{-(-2)} \\ &= 7.389 \end{aligned}$$